

ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 2

Data Quality, Ethics & Professional Practice



Slides and interactive demos (click or scan):
aardid.github.io/uc_205_gis_course

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Engineers make decisions, not just maps.

The quality of your decision depends entirely on the quality of your data.

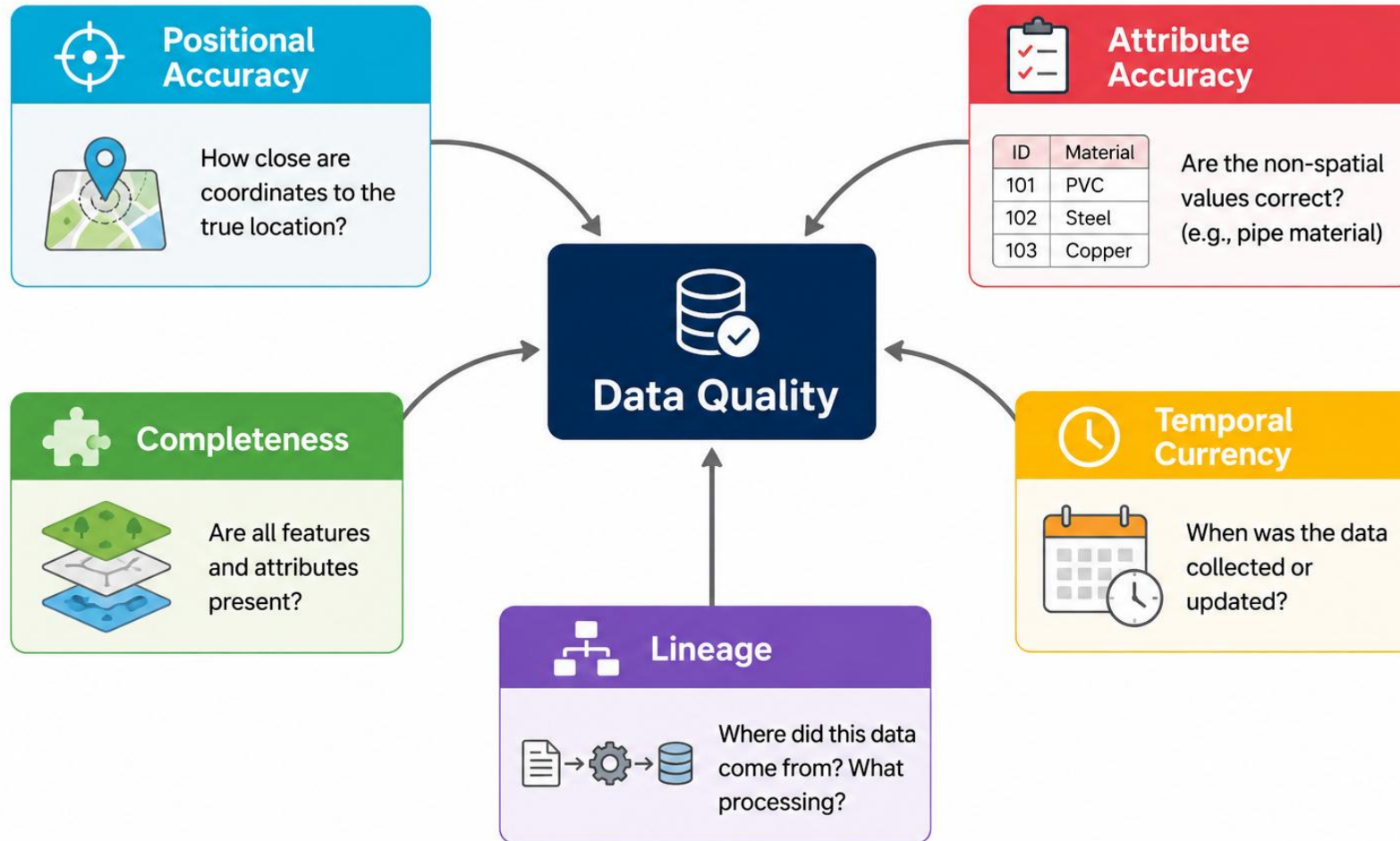
What happens when the GPS coordinates are 5 metres off?

What happens when a pipe is recorded as concrete but is actually asbestos?

What happens when the flood zone boundary was mapped 15 years ago?

Today: how to assess, communicate, and work ethically with spatial data.

The Five Dimensions of Spatial Data Quality



Every spatial dataset should be assessed across all five dimensions before use.

Positional Accuracy

How close are recorded coordinates to the true location?

Consumer GPS: $\pm 3\text{--}5$ m typical

Survey-grade GNSS: $\pm 1\text{--}2$ cm

Digitised from aerial imagery: depends on resolution

Historical maps: may be $\pm 10\text{--}50$ m

For pipe location: 5 m error could mean digging in the wrong place.

For regional planning: 5 m error is negligible.



GPS surveying equipment

Always ask: 'how accurate — and how do you know?'

Attribute Accuracy & Completeness

Attribute Accuracy

Are the non-spatial values correct?

Pipe recorded as 'concrete' but is actually 'asbestos cement'

Building height entered as 3.5 m instead of 35 m (decimal error)

Land use class changed since mapping (pastoral → residential)

Completeness

Are all features and attributes present?

Only 60% of pipes have material recorded

Rural buildings not digitised (urban bias)

Entire suburbs missing from older datasets

Missing data ≠ 'no data exists' — it may mean 'never collected'.

Always ask: 'what is missing from this dataset?'

Temporal Currency & Lineage

Temporal Currency

When was the data collected or last updated?

A flood model using 2008 terrain is missing post-earthquake ground changes

A building footprint layer from 2015 misses new subdivisions

Census data is collected every 5 years — what happened since?

Lineage (Provenance)

Where did this data come from? What processing was done?

Who collected it? (LINZ, council, consultant, volunteer?)

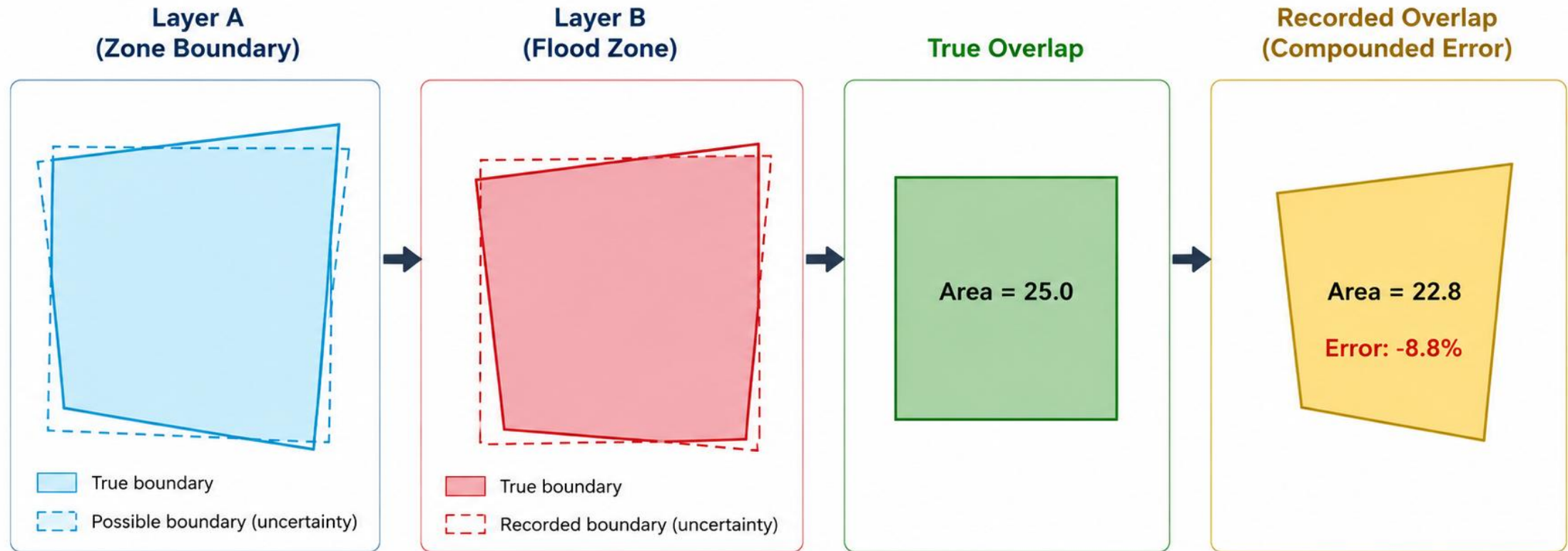
What method? (survey, digitisation, modelling, crowd-source?)

What transformations? (reprojected, clipped, simplified?)

Good metadata documents the lineage. No metadata = unknown quality.

Always ask: 'when was this collected - and by whom?'

Error Propagation - Errors Compound Through Analysis



Key Point: Uncertainty in input layer boundaries propagates through spatial overlay operations, resulting in compounded area (or attribute) errors.

Small errors in input layers combine to create larger errors in the result.

This is why fitness-for-purpose assessment matters BEFORE analysis, not after.

The Fitness-for-Purpose Test

Is this data good enough for THIS decision?

There is no universal standard -- quality is relative to the use case:

Regional flood mapping: ± 5 m positional accuracy is acceptable

Pipe trenching: ± 5 m means digging in the wrong spot

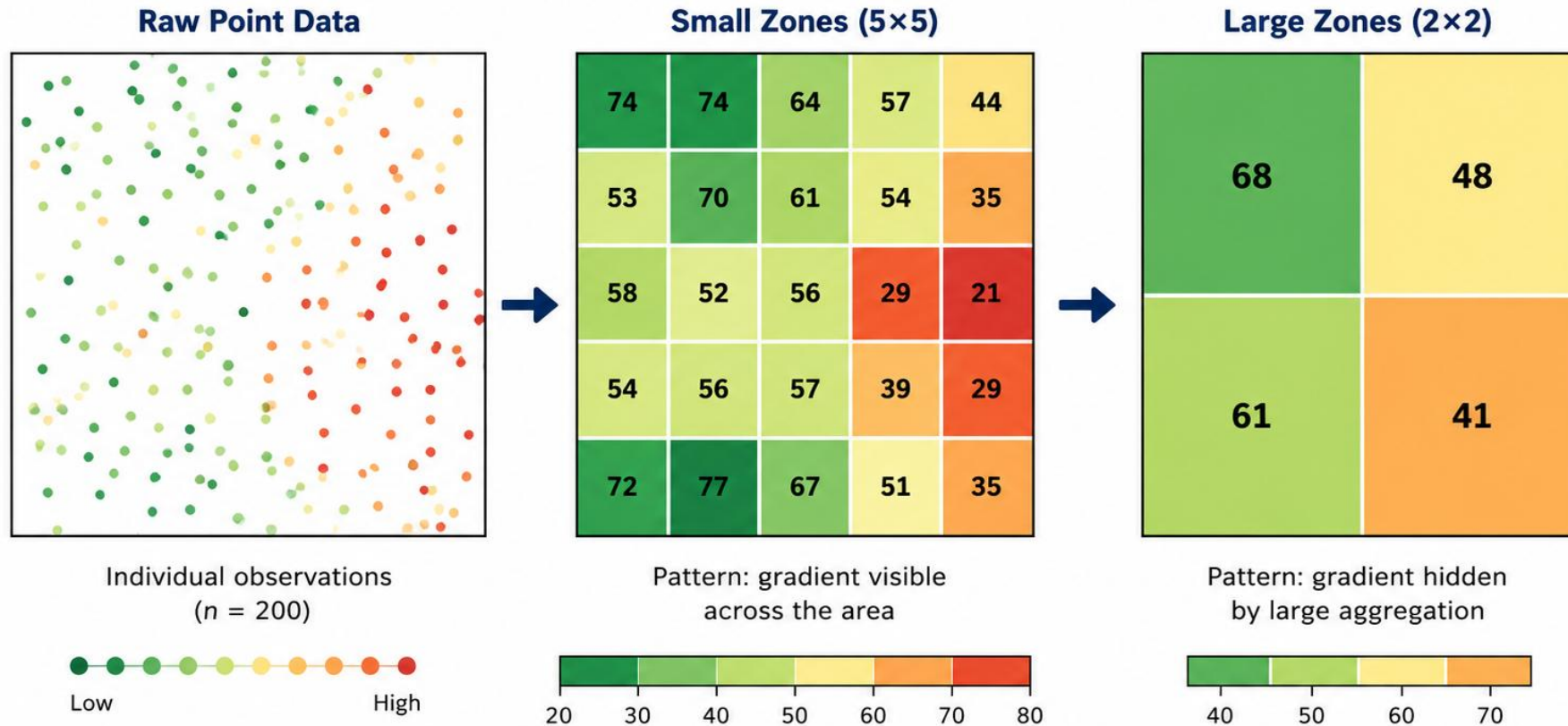
Census data from 2018: fine for 2020 planning, risky for 2026

Volunteer-mapped buildings: useful for overview, not for structural assessment

The professional engineer's job: assess fitness-for-purpose before using data.

Document your assessment and communicate limitations clearly.

The Modifiable Areal Unit Problem (MAUP)



Same data, different zone boundaries → different spatial patterns and conclusions.

The zones you choose determine the patterns you see.

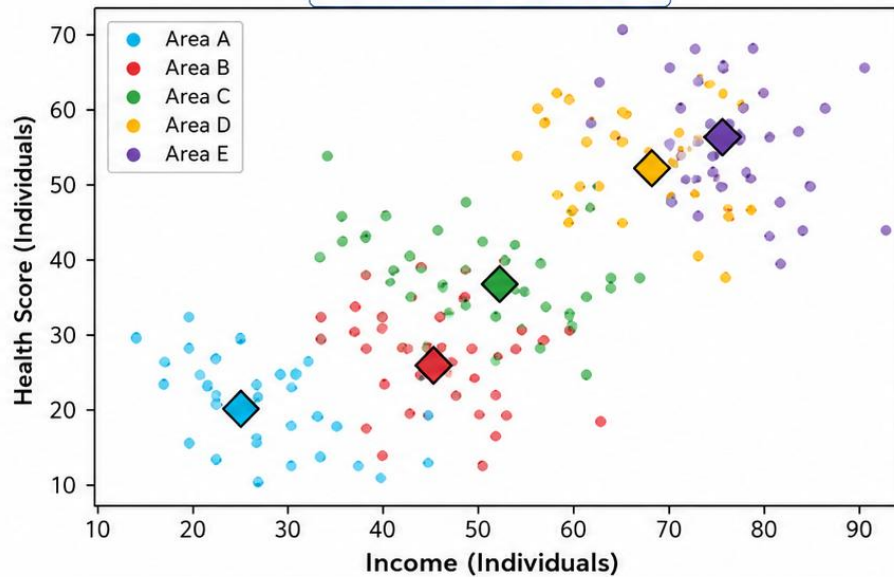
Different zone boundaries applied to the same data can lead to opposite conclusions.

Always ask: 'what zones were used?'

The Ecological Fallacy

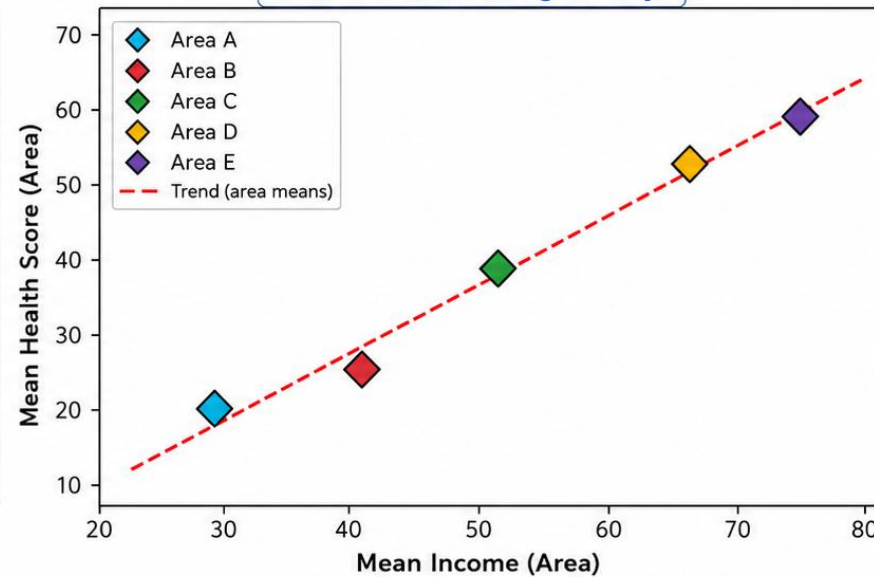
Ecological Fallacy — Area Trends Can Mislead About Individuals

Individual-Level Data



Within each area: wide variation in individual health scores at similar income levels.
Individual-level relationships are noisy.

Area-Level Averages Only

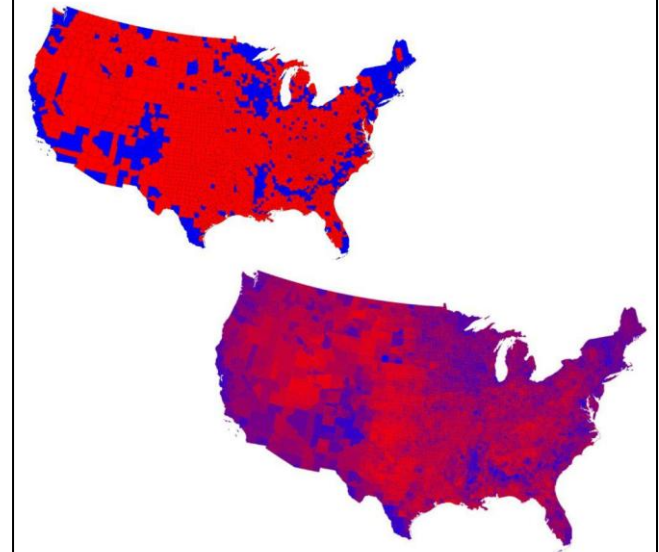


Between areas only: strong positive relationship appears when using area averages.
Aggregated trends can be misleading about individuals.



Key Point: Do not infer individual-level relationships from area-level (aggregate) data.
This is the **ecological fallacy**.

US Presidential Election Results by County (2012)



You cannot infer individual behaviour from area-level averages.

The trend between areas does not mean the same trend exists within areas.

Always ask: 'at what level was this measured?'

Data Ethics — Location Data is Personal Data

NZ Privacy Act 2020

Location data can identify individuals — even without names.

A home address + time-stamped GPS trace = personal information.

Key principles for engineers:

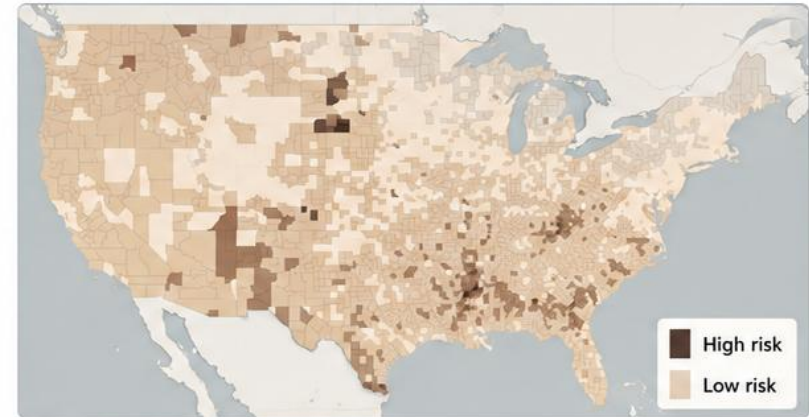
Collect only what you need (data minimisation)

Store securely and limit access

Anonymise or aggregate before sharing

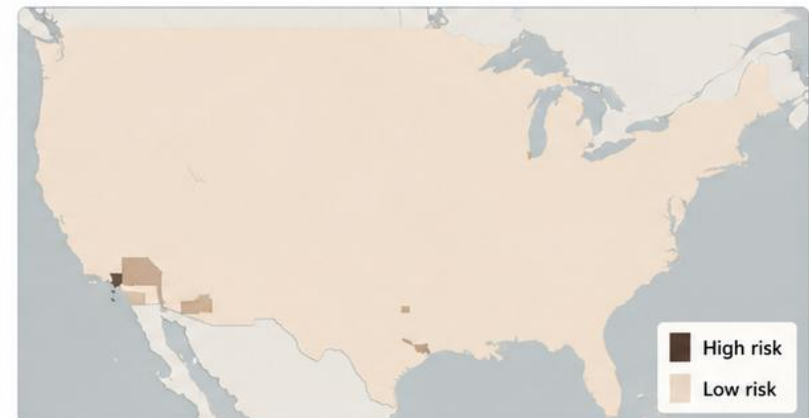
Be transparent about what you collect and why

Example: Publishing a map of 'flood-risk properties' without owner consent may violate the Privacy Act.



Detailed data: many small areas

More precise but higher risk of re-identification



Aggregated data: fewer large areas

Less precise but lower risk of re-identification



Respect privacy. Protect people. Build trust. Ethical data practices benefit everyone.

Te Mana Raraunga -- Māori Data Sovereignty

Data about Māori should be governed by Māori.

Te Mana Raraunga (Māori Data Sovereignty Network) principles:

Rangatiratanga — Māori have authority over Māori data

Whakapapa — data has origins and connections, like people

Whanaungatanga — data should serve relationships and communities

Kaitiakitanga — responsible guardianship of data

Relevance for engineers:

When working with spatial data about Māori land, resources, or communities, you must engage with iwi/hapū and respect data governance protocols.

This includes environmental data on land of significance to Māori.



LIVE DEMO

[L02_DataQuality.html](#)

Visualise how GPS positional error affects buffer and overlay results.
Toggle error magnitude to see the impact on engineering decisions.

Scan to open the demo on your phone



aardid.github.io/uc_205_gis_course/L02_DataQuality.html
All slides & demos: aardid.github.io/uc_205_gis_course

Professional Practice -- What NZ Employers Expect

Data quality is a professional competency, not a GIS technicality.

NZ employers (councils, consultancies, Crown entities) expect engineers to:

Assess data quality before using it — don't trust blindly

Document data sources and limitations in reports

Communicate uncertainty to clients and decision-makers

Follow privacy and ethical requirements

Engineering NZ Code of Ethical Conduct:

"Take reasonable steps to safeguard health and safety" — this includes data-driven decisions.

"Be honest and objective" — communicate data limitations, don't hide uncertainty.

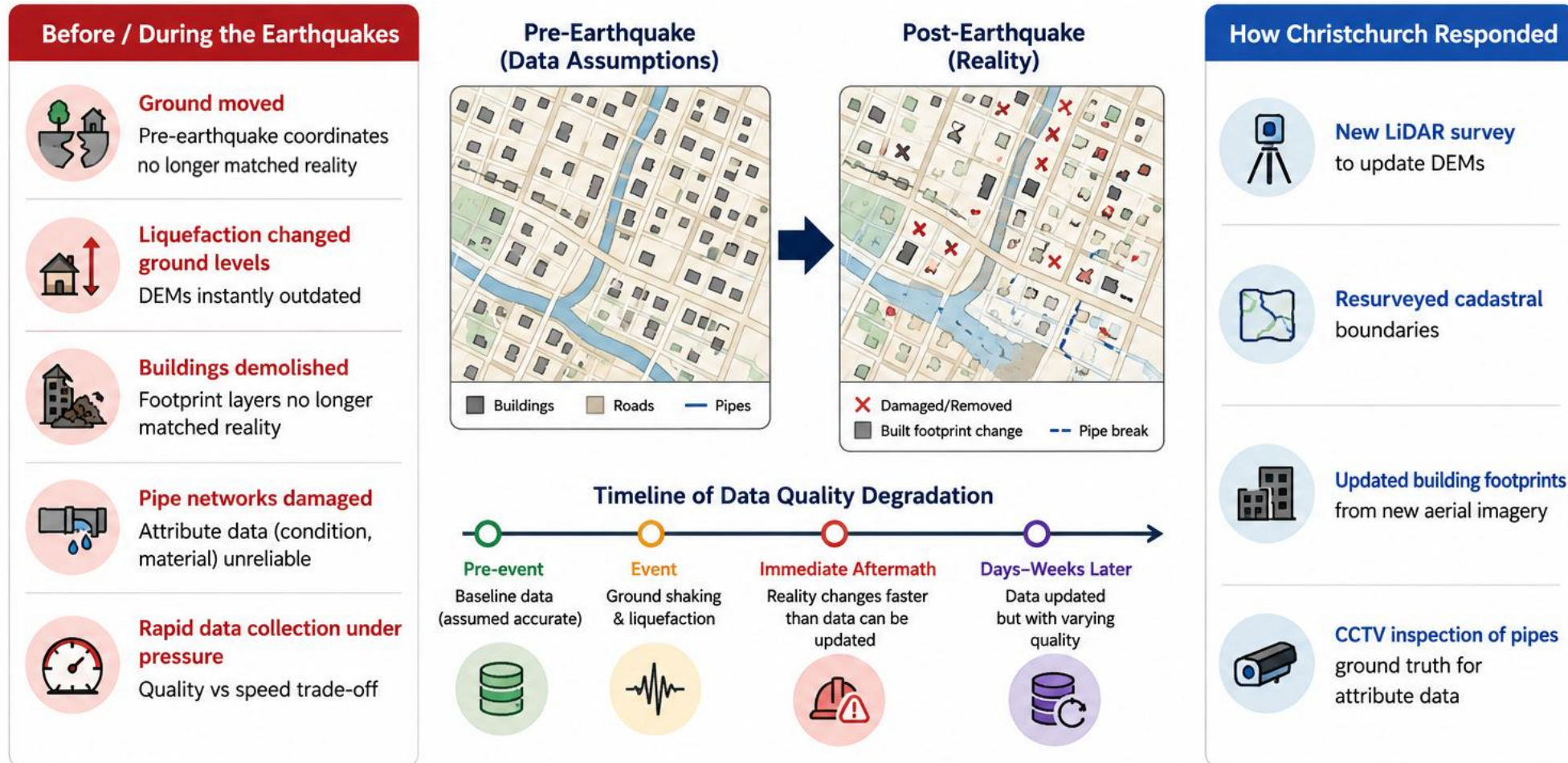
The Data Quality Checklist

Before ANY spatial analysis, check:

1. Source — Who created this data? Is the provider reputable?
2. Date — When was it collected/updated? Is it current enough?
3. Accuracy — What positional and attribute accuracy is documented?
4. Completeness — Are features or attributes missing? Where are the gaps?
5. CRS — What coordinate reference system? Is it consistent with other layers?
6. Metadata — Is there documentation? Can you trace the lineage?
7. Fitness — Is this data good enough for YOUR specific analysis?

If you can't answer these questions, you can't trust your results.

Canterbury Case Study -- Earthquake Data Quality Challenges



Lesson: Data quality is not static — events can invalidate everything.

KEY CONCEPTS

Today's Takeaways

1. Data quality has five dimensions: positional accuracy, attribute accuracy, completeness, temporal currency, lineage
2. Errors propagate and compound through spatial analysis — check BEFORE, not after
3. MAUP and ecological fallacy are traps: aggregation choices change conclusions
4. Fitness-for-purpose: quality is relative to the decision you're making
5. Ethics matter: NZ Privacy Act, Te Mana Raraunga, and the Engineering NZ Code apply to spatial data

NEXT LECTURE

Lecture 3: Spatial Data Models

Canterbury Case Study — Earthquake Data Quality Challenges

The 2010-2011 earthquakes created massive data quality challenges:

Ground moved — pre-earthquake coordinates were wrong

Liquefaction changed ground levels — DEMs instantly outdated

Buildings demolished — footprint layers no longer matched reality

Pipe networks damaged — attribute data (condition, material) unreliable

Rapid data collection under pressure — quality vs speed trade-off

How Christchurch responded:

New LiDAR survey to update DEMs

Resurveyed cadastral boundaries

Updated building footprints from new aerial imagery

CCTV inspection of pipes — ground truth for attribute data

Lesson: data quality is not static — events can invalidate everything.

Common Mistakes to Avoid

Assuming all data is accurate because it comes from a 'reputable' source

Not checking the CRS — mixing NZTM with WGS84 causes silent errors

Ignoring temporal currency — using 10-year-old data for current decisions

Presenting results without documenting data limitations

Inferring individual behaviour from area-level data (ecological fallacy)

Choosing aggregation zones that support your hypothesis (MAUP bias)

The antidote: apply the data quality checklist before every analysis.

**You receive a dataset from a client with no metadata.
How do you decide whether to use it?**

Consider:

What questions would you ask the client?

What checks can you run on the data itself?

How would you document your quality assessment?

At what point would you refuse to use it?